Theory of Computation

BSc Computer Science - Durham University

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1 Module Introduction

Theory of Computation has the module code COMP2181 and is led by Dr Barnaby Martin.

1.1 Assessment

This module is assessed through two summative benchtests, and one final exams, these can be seen in Table 1. Additional formatives are also shown in italics.

Title	Opening	Due	Feedback	Professor	Weight
Benchtest 1	Week 11	-	2nd Feb	Dr Dantchev and Dr Mertzios	12.5%
Benchtest 2	Week 20	-	-	Dr Dantchev and Dr Mertzios	12.5%
Final Exam	May Exam	-	-	Dr Barnaby Martin	75%

Table 1: Assessments for Theory of Computation.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Tuesday	10:00 - 11:00	Lecture	CLC202	Dr Barnaby Martin
Wednesday	09:00 - 11:00	Problem Class	TLC116	-
Thursday	14:00 - 15:00	Lecture	CLC202	Dr Sukanya Pandey

Table 2: Michaelmas Term Timetable for Theory of Computation.

2 Models of Computation

This section is taught by Dr Barnaby Martin. Check for recommended reading.

2.1 Finite-State Automata and Regular Languages

2.1.1 Formal Definitions

Definition 2.1.1.1: A deterministic finite-state automaton is a five-tuple $(Q, \Sigma, \delta, q_0, F)$ where Q is the finite set of states, Σ is a finite alphabet, $\delta : Q \times \Sigma \to Q$ is the transition function, $q_0 \in Q$ is the start state, and $F \subseteq Q$ is the set of accept states.

Definition 2.1.1.2: A DFA M recognises the language L if $L = \{w | M \text{ accepts } W\}$.

Definition 2.1.1.3: A language is called a regular language if some DFA recognises it.

Definition 2.1.1.4: A regular expression R defines a language $L(R)$. We shall eventually prove that $RE \equiv DFA$. R is a regular expression over the alphabet Σ if R is:

1. a for some $a \in \Sigma$.
2. ϵ (the empty string).
3. $\emptyset$ (the empty set).
4. $(R_1 \cup R_2)$ where R_1 and R_2 are regular expressions.
5. $R_1 \cdot R_2$ where R_1 and R_2 are regular expressions.
6. R_1^* where R_1 is a regular expression.

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The list of regular operations can be seen in Table 3 where the operations labelled with $^+$ are language theory specific.

Operation Name	Notation	Set Equivalent
Union	$A \cup B$	$\{x \mid x \in A \text{ or } x \in B\}$
Intersection	$A \cap B$	$\{x \mid x \in A \text{ and } x \in B\}$
Difference	$A \backslash B$	$\{x \mid x \in A \text{ and } x \notin B\}$
Complement	$\overline{A}$	$\Sigma * \backslash A$
Concatenation$^+$	$A \circ B$	$\{xy \mid x \in A \text{ and } y \in B\}$
Star$^+$	$A*$	$\{x_1 x_2 ... x_k \mid k \geq 0 \text{ and } x_i \in A \text{ for every } i, 1 \leq i \leq k\} \equiv \cup_{i=0}^{\infty} A^i$

Table 3: Regular Operations.

2.1.2 Combining Automata

Sometimes we may want to implement an operation on multiple languages. We can do this by combining their automata. If we have that L_1 is recognised by $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ and L_2 is recognised by $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$; we want to combine M_1 and M_2 into a new automaton M that would recognise $L_1 \cup L_2$.

You would think that we could just run the automata sequentially, and accept if either M_1 or M_2 accept, but this does not work. This is because after M_1 has run on the input, the input is exhausted and there is no way to rewind it in order to run M_2.

Instead, we can attempt to run M_1 and M_2 on the input in parallel. We do this by using pairs of states (one for each DFA). Formally, the components of the new automaton $M = (Q, \Sigma, \delta, q_0, F)$ are:

1. The set of possible states contains every possible cross-set pair of Q_1 and Q_2.

$$Q = Q_1 \times Q_2$$

2. For every $s \in Q_1, u \in Q_2, a \in \Sigma$. The transition function contains every transition from a pair of states from Q being given any input from Σ.

$$\delta((s, u), a) = (\delta_1(s, a), \delta(u, a))$$

3. The start state is the pair containing both start states from M_1 and M_2.

$$q_0 = (q_1, q_2)$$

4. The set of accept states is every possible cross-set pair of accept states from F_1 and F_2.

$$F = \{(s, u) \in Q \mid s \in F_1 \text{ or } u \in F_2\}$$

2.1.3 Non-Deterministic Finite Automaton

If, instead you wish to work with concatenation, such that $w \in L_1 \circ L_2$ only if there are words x and y such that $w = xy$, $x \in L_1$ and $y \in L_2$. You need to run M_1 on a prefix of w and then run M_2 on the rest. This must be done non-deterministically and stops becoming a DFA as all accept states of M_1 have to map to the start state of M_2.

Definition 2.1.3.1: A non-deterministic finite automaton is a five-tuple $(Q, \Sigma, \delta, q_0, F)$, where Q is the finite set of states, Σ is a finite alphabet, $\delta : Q \times (\Sigma \cup \{\epsilon\}) \to \mathscr{P}(Q)$ is the transition function, $q_0 \in Q$ is the start state, and $F \subseteq Q$ is the set of accept states.

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Computing NFA

Before computing an NFA, an important convention to note is that a single step in a computation consists of first reading the next input symbol followed by taking any number of ϵ-transitions. This means that empty strings ϵs can be freely added inside the actual word $\in \Sigma*$, e.g., 010110 can be viewed as, say, $\epsilon 01\epsilon\epsilon 01\epsilon 01$.

Let $M = (Q, \Sigma, \delta, q_0, F)$ be an NFA and $w = w_1 w_2 ... w_n$ be a word over $\Sigma \cup \{\epsilon\}$. M accepts w if there exists a sequence of states $r_0, r_1, r_2, ..., r_n$ satisfying the following conditions:

1. $r_0 = q_0$.
2. $r_{i+1} \in \delta(r_i, w_{i+1})$ for every $i, 0 \leq i \leq n-1$.
3. $r_n \in F$.

2.1.4 Converting NFA into DFA

The difference between a DFA and NFA is that a DFA has a unique computation (path) and either accepts or rejects, while an NFA generally has many computation paths and accepts if at least one computation accepts and only rejects if all the computations reject.

Theorem 2.1.4.1: Every NFA has an equivalent DFA.

Proof: To formally convert an NFA to a DFA, let $N = (Q, \Sigma, \delta, q_0, F)$ be an NFA recognising some language L (assume that N has no ϵ-transitions). We shall construct a DFA $M = (Q', \Sigma, \delta', q_0', F')$ that recognises the same language L as follows.

1. $Q' = \mathscr{P}(Q)$.
2. For $R \in Q'$ and $a \in \Sigma$, let

$$\delta'(R, a) = \{q \in Q | q \in \delta(r, a) \text{ for some } r \in R\} \tag{1}$$

   equivalently,

$$\delta'(R, a) = \bigcup_{r \in R} \delta(r, a) \tag{2}$$

3. $q_0' = \{q_0\}$.
4. $F' = \{R \in Q' | \exists r \in R, r \in F\}$.

Now, to consider ϵ-transitions, for $R \subseteq Q$, let

$$E(R) = \{q | q \text{ is reachable from } R \text{ through a number of } \epsilon\text{-transitions.}\} \tag{3}$$

Now, taking the above steps, we modify (1) to

$$\delta'(R, a) = \{q \in Q | q \in E(\delta(r, a)) \text{ for some } r \in R\} \tag{4}$$

and (2) to

$$\delta'(R, a) = \bigcup_{r \in R} E(\delta(r, a)) \tag{5}$$

We then modify step 3 to say $q_0' = E(\{q_0\})$.

□

Corollary 2.1.4.1: A language is regular iff some NFA recognises it.

Proof: See Theorem 2.1.4.1.

□

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Theorem 2.1.4.2: A language is regular iff some regular expression describes it.

Proof: $\Leftarrow$ Given a regular expression R, construct a NFA that recognises $L(R)$ by structural induction:

- *Base Case*: All initial regular expressions can be implemented by NFAs.
- *Inductive Step*: We have already proven that the class of regular languages is closed under regular operations.

$\Rightarrow$ Given a DFA M, construct a regular expression that recognises $L(M)$.

1. Convert the given n-state DFA into an equivalent $n+2$ state GNFA.
2. Eliminate all the internal states of the GNFA one-by-one, while preserving the language recognised by the automaton.
   (a) Pick an internal state r for elimination.
   (b) For every two states u and v with transitions between them and r, $u \neq r, v \neq r$, combine the regular expressions of the three into one, new expression.
3. $L(R)$ should now equal $L(M)$.

□

Definition 2.1.4.1: A generalised non-deterministic finite automaton is defined as having a single start state with no incoming edges, a single end state with no outgoing edges, and every transition being labelled with a regular expression.

2.1.5 Non-Regular Expression

Definition 2.1.5.1: A non-regular expression is a language which cannot be recognised by any DFA. We prove a language is non-regular by assuming there is a DFA that recognises language L and then use an adversary argument against that DFA to get a contradiction.

Lemma 2.1.5.1: The pumping lemma states that, for every regular language L, there is a number p (the pumping length of L) such that every word $w \in L$ of length at least p can be divided into three parts $w = xyz$, satisfying the following conditions:

1. $xy^iz \in L$ for every $i \in N$ (in particular $xz \in L$ as one can take $i = 0$).
2. y is a non-empty string.
3. The length of the string xy is not greater than p.

Proof: Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA that recognises L, and let us set the pumping length p to be the number of states in $M : p = |Q|$. Consider the computation of M on the word $w = w_1w_2...w_n$ where $n \geq p$.

$$q_0 = r_0 \rightarrow w_1 \rightarrow r_1 \rightarrow w_2 \rightarrow ...r_{n-1} \rightarrow w_n \rightarrow r_n \in F \tag{6}$$

Let i be the first moment in time when we see a repetition in the sequence of states $r_0, r_1, ..., r_n$ (such an i must exist by the Pigeon-Hole principle, as $n+1 > p$). i.e., $r_i = r_j$ for some $i > j$ and the states $r_0, r_1, ..., r_{i-1}$ are all different. We can then set

$$x = w_1w_2...w_j \quad y = w_{j+1}...w_i \quad z = w_{i+1}...w_n \tag{7}$$

It is now straightforward to check that all three conditions are met.

□

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Example 2.1.5.1: Prove that the language $L = \{0^n 1^n | n \in \mathbb{N}\}$ is not regular.

Proof: Take the word $0^p 1^p \in L$ where p is the pumping length of L. If $0^p 1^p = xyz$ and $|xy| \leq p$, then $x = 0^i, y = 1^j$ and $z = 0^{p-i-j}1^p$. But then $xy^2z = 0^{p+j}1 \notin L$ as $|y| > 0$, i.e., $j > 0$.

□

Definition 2.1.5.2: An algorithm that minimises any given DFA, i.e., finds an equivalent DFA with the minimal number of states. It answers both "What is the smallest DFA that recognises the same language?" and "Given two DFAs, do they recognise the same language?". It can be achieved by removing any states not reachable from the start state and contracting a set of states that are equivalent into a single state.

Definition 2.1.5.3: Two states s and t are equivalent if any word w, which is accepted when starting from s, is also accepted when starting from t.

Definition 2.1.5.4: A extended transition function is defined by letting a DFA $M = (Q, \Sigma, \delta, q_0, F)$ be given, and let $\hat{\delta} : Q \times \Sigma^* \to Q$ be defined as:

$$\hat{\delta}(s, \epsilon) = s \tag{8}$$

$$\hat{\delta}(s, w_1 w_2 ... w_n) = \hat{\delta}(\delta(s, w_1), w_2, ..., w_n) \tag{9}$$

for every state s and every word $w_1 w_2 ... w_n$.

2.2 Büchi Automata and Omega-Regular Languages

2.2.1 Finite-State Automata Over Infinite Words

Definition 2.2.1.1: If we have a NFA where the input is infinite (i.e., Σ^ω), we have a Büchi automaton. A Büchi automaton accepts $w_1 w_2 ... \in \Sigma^\omega$ if there is a sequence of states $r_0, r_1, r_2, ... \in Q^\omega$ satisfying the following conditions:

- $r_0 = q_0$.
- $(r_i, w_{i+1}, r_{i+1}) \in \Delta$ for every $i > 0$.
- There are infinitely many r_i's in F.

Definition 2.2.1.2: An ω-regular language/expression is build upon regular languages, using the following operations:

- $A \cup B$, where both A and B are ω-regular.
- AB, where A is regular, and B is ω-regular.
- A^ω, which is $\{a_1 ... | a_i \in A\}$, i.e., an infinite sequence of words from A, where A is regular and doesn't contain the empty word.

Let A be a regular language. The limit of A is the language $\{a \in \Sigma^\omega | a \text{ has infinitely many prefixes in } A\}$.

$L_1 = \{a \in \Sigma^\omega | a \text{ contains infinitely many 0s.}\}$.
$L_2 = \{a \in \Sigma^\omega | a \text{ contains finitely many 1s.}\}$.

Proposition 2.2.1.1: L_2 cannot be recognised by a deterministic Büchi automaton.

Proof: Assume that L_2 can be recognised by some deterministic Büchi automaton M. We will construct a word accepted by M that contains infinitely many 1s and thus doesn't belong to L_2.

1. $0^\omega \in L_2$ must be accepted by M, so there is some finite prefix 0^{n_1} that leads into accepting state p_1.
2. $0^{n_1}10^\omega \in L_2$ must be accepted by M, so there is some finite prefix $0^{n_1}10^{n_2}$ that leads into accepting state p_2.

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3. $0^{n_1}10^{n_2}10^{\omega} \in L_2$ must be accepted by M, so there is some finite prefix $0^{n_1}10^{n_2}10^{n_3}$ that leads into accepting state p_3.

4. ...

The limit of the infinite process is a word of the form $(0^*1)^{\omega} \notin L_2$ that is accepted by M since it visits an infinite sequence of accepting states $p_1, p_2, ...$

□

2.2.2 Modal Logic

Definition 2.2.2.1: The box operator is used with propositional logic $\Box\varphi$ to mean "I know/can prove" φ. More generally - necessarily φ. Probably $\Box\varphi \rightarrow \varphi$, but not the other way round (φ is true, but we may not be able to prove it).

Definition 2.2.2.2: The diamond operator is used in $\neg\Box\varphi \stackrel{\text{def}}{=} \diamond\neg\varphi$ to indicate that you cannot prove φ, so it is possible that $\neg\varphi$. Conversely, $\neg \diamond \varphi$ (φ is impossible) is the same as proving $\neg\varphi$, i.e., $\Box\neg\varphi$. Simply, $\diamond$ means "it is possible that".

From these definitions and propositional logic, we get:

- $\frac{\varphi}{\Box\varphi}$ is the Neccesitation Rule **N**.
- $\Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi)$ is the Distribution Axiom **K**.

Definition 2.2.2.3: Kripke semantics state that, over a set of atomic propositions (Boolean variables) P, and a set of worlds (states) W, we are given a valuation function

$$f : P \times W \rightarrow \{true, false\} \tag{10}$$

and an accessibilty relation

$$R \subseteq W \times W \tag{11}$$

$(u, v) \in R$ if v is accessible from u. $\Box a$ holds at u iff a holds at every v, s.t. $(u, v) \in R$. Thus, $\diamond a$ holds at u iff a holds at some v, s.t. $(u, v) \in R$.

2.3 Linear-Time Logic and Translating Into Büchi Automata

3 Algorithms and Complexity

This section is taught by Dr Sukanya Pandey. The recommended reading for this topic is:

- Introduction to Algorithms (Cormen T.H., et al., 2009).

3.1 Asymptotic Complexity and Notation

Lecture one was just a recall of some of last year's Algorithms and Data Structures module.

3.2 Greedy Algorithms

Definition 3.2.0.1: The knapsack problem is defined as one of two variants. Either:

- The 0-1 knapsack problem (the classic variant), where a thief robs a store and finds n items. Item i is worth v_i euros and weighs w_i kilos. The thief can carry at most W kilos in his knapsack. *Which items can he take to maximise his profit?*

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- the fractional knap problem, which has the same setup, but this time the thief can take fractions of items.
- We can assume (without loss of generality) that the knapsack is not big enough to carry all items:

$$W < w_1 + w_2 + ... + w_n \tag{12}$$

3.2.1 The Greedy Approach

To solve the fractional problem, the greedy approach would be to calculate the value per kilo using (13) for each item. You then sort the items by value per kilo, in decreasing order. You then take as much of the first item as possible, then the second item, etc., until you reach the capacity of the knapsack W.

$$r_i = \frac{v_i}{w_i} \tag{13}$$

Example 3.2.1.1: For example, if $W = 10$, and $\mathbf{v} = \{4, 10, 12, 20\}$ and $\mathbf{w} = \{4, 2, 8, 5\}$. We can then calculate $\mathbf{r} = \{1, 5, 1.5, 4\}$. The greedy algorithm then dictates that you should take all of item 2, all of item 4, and 3 kilos of item 3 $W = 2 + 5 + 3$. This gives a total value of $V = 10 + 20 + (3 \times 1.5) = 34.5$.

The running time of this algorithm can be calculated by taking that:

1. Step 1, calculating the value per kilo takes $\Theta(n)$.
2. Step 2, sorting the value per kilo array takes $\Theta(n\ log\ n)$.
3. Step 3, selecting the items can be done in $\Theta(n)$.

Therefore, the worst-case running time is $\Theta(n\ log\ n)$.

However, approach will not work for the 0-1 knapsack problem.

3.2.2 Correctness of the Greedy Algorithm for Fractional Knapsack

n items are sorted by value per kilo, (13) in decreasing order, i.e.,

$$r_1 \geq r_2 \geq ... \geq r_n \tag{14}$$

Two items, p and q with $r_p = r_q$ are considered to be equivalent. Hence, we may assume that

$$r_1 > r_2 > ... > r_n \tag{15}$$

The total available amount of item i is w_i. Let k be such that

$$w_1 + w_2 + ... + w_k \leq W \tag{16}$$

and

$$w_1 + w_2 + ... + w_{k+1} > W \tag{17}$$

Let $\mathbf{x} = (x_1, x_2, ..., x_n)$ denote a solution: $0 \leq x_i \leq w_i$ is the amount the thief takes of item i for $i = [0, n]$. The total value taken is then given by

$$\sum_{i=1}^{n} r_i x_i \tag{18}$$

The greedy solution $\mathbf{x}$ is given by $x_i = w_i$ for $1 \leq i \leq k$ and

$$x_{k+1} = W - (w_1 + w_2 + ... + w_k) \tag{19}$$

$x_i = 0$ for $k + 2 \leq i \leq n$. Let $\mathbf{x}^* = (x_1^*, x_2^*, ..., x_n^*)$ be an optimal solution.

Suppose that $\mathbf{x}^* \neq \mathbf{x}$. Then

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- $x_i^* < x_i$ for some $i \leq k+1$.
- Let $j = \min\{i : x_i^* < x_i\}$.
- And $x_l^* > x_l$ for some $l > j$.

Let us add some more of item j to the optimal choice. Let y be another solution, defined by

$$y = \begin{cases} y_j = \min\{w_j, x_j^* + (x_l^* - x_l)\} \\ y_l = x_l^* - (y_j - x_j^*) \\ y_i = x_i^* \text{ otherwise.} \end{cases} \tag{20}$$

3.3 Divide-and-Conquer Algorithms

3.3.1 Colouring for Restricted Inputs

Definition 3.3.1.1: Many useful algorithms utilise recursions to quickly resolve subproblems. We refer to these as divide-and-conquer algorithms, as at each step they:

1. *Divide* the problem into a number of subproblems.
2. *Conquer* the subproblems by solving them, either recursively or directly depending on size.
3. *Combine* the solutions for the subproblems into a solution for the original problem.

A natural way to calculate the running time of these algorithms is recurrence relations using the Master Theorem (covered in Year 1 - Algorithms and Data Structures).

Definition 3.3.1.2: A colouring of a graph is an assignment of colours to its vertices such that any two adjacent vertices get a different colour. A colouring using at most k colours is a k-colouring.

Definition 3.3.1.3: The chromatic number, χ_G, of a graph is the smallest integer k for which G has a k-colouring. Calculating the chromatic number of any given graph is considered a NP-hard problem.

Definition 3.3.1.4: Let G_1 and G_2 be two vertex-disjoint graphs. The disjoint union $G_1 + G_2$ is the graph with vertex set $V(G_1) \cup V(G_2)$ and edge set $E(G_1) \cup E(G_2)$.

Definition 3.3.1.5: Let G_1 and G_2 be two vertex-disjoint graphs. The join $G_1 \times G_2$ is the graph with vertex set $V(G_1) \cup V(G_2)$ and edge set $E(G_1) \cup E(G_2) \cup E^*$, where E^* is the set of all edges between vertices of G_1 and vertices of G_2.

Definition 3.3.1.6: Let G_1 and G_2 be two vertex-disjoint graphs. A graph G is a cograph if it can be created using the following rules:

- The 1-vertex graph is a cograph.
- If G_1 and G_2 are cographs, then so is $G_1 + G_2$.
- If G_1 and G_2 are cographs, then so is $G_1 \times G_2$.

Alternatively, let P_4 denote the path on 4 vertices. A graph G contains P_4 as an induced subgraph if G can be modified to P_4 via vertex deletions. If not, then G is said to be P_4-free.

Theorem 3.3.1.1: A graph G is a cograph $\iff$ G is P_4-free.
Proof: Proof excluded.

□

Definition 3.3.1.7: Let G be a cograph. The cotree T_G of G is the unique decomposition tree that satisfies:

1. The root r of the tree represents the graph $G_r = G$.

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2. Every leaf x of the tree represents exactly one vertex of G, and vice-versa.
3. Every internal node x of the tree has at least two children and is either labelled + or $\times$. The node represents an induced subgraph G_x of G, defined as follows:
   - If x is a +-node, then G_x is the disjoint union of all graphs G_y where y is a child of x.
   - If x is a $\times$-node, then G_x is the join of all graphs G_y where y is a child of x.
4. Along the path from the root r to any leaf, the labels of the internal notes alternate between + and $\times$.

Theorem 3.3.1.2: Any cograph G has a unique cotree. However, if we drop condition 4, there may be many trees representing G.

Proof: Proof excluded.

□

We can use a divide-and-conquer algorithm to colour a cograph. Given a cograph G, construct its cotree T_G. This can be done in $O(n+m)$ time (Corneil, Perl and Stewart, 1985). We follow a bottom-up approach: only process node x after first processing its children.

- *Leaves*: Each leaf represents a single vertex of G. The chromatic number of a single-vertex graph is 1.
- *Union-Nodes*: If x is a +-node, then χ_{G_x} is the maximum of the chromatic numbers of the graphs G_y, where y is a child of x.
- *Join-Nodes*: If x is a $\times$-node, then χ_{G_x} is the sum of the chromatic numbers of the graphs G_y, where y is a child of x.

This way, the last node we process is the root r, which represents the graph $G_r = G$. As the number of nodes of T is $O(n)$ and we used $O(n)$ time per node, the total running time is polynomial.

Algorithm 1 MAX-SUBARR

Input: A, low, high
Output: low, high, sum

if high = low **then return** (low, high, A[low])
else
  median $\leftarrow \left\lfloor \frac{\text{low+high}}{2} \right\rfloor$
  $(L_{low}, L_{high}, L_{sum}) \leftarrow$ MAX-SUBARR(A, low, median)
  $(R_{low}, R_{high}, R_{sum}) \leftarrow$ MAX-SUBARR(A, median+1, high)
  $(\text{Cross}_{low}, \text{Cross}_{high}, \text{Cross}_{sum}) \leftarrow$ MAX-CROSS-SUBARR(A, low, median, high)
  if $L_{sum} \geq \max\{R_{sum}, \text{Cross}_{sum}\}$ **then return** $(L_{low}, L_{high}, L_{sum})$
  end if
  if $R_{sum} \geq \max\{L_{sum}, \text{Cross}_{sum}\}$ **then return** $(R_{low}, R_{high}, R_{sum})$
  else return $(\text{Cross}_{low}, \text{Cross}_{high}, \text{Cross}_{sum})$
  end if
end if

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Algorithm 2 MAX-CROSS-SUBARR

Input: A, low, mid, high
Output: max_left, max_right, left_sum+right_sum

```
left_sum← −∞
for i ←mid to low do
    sum ← sum+A[i]
    if sum > left_sum then
        left_sum ← sum
        max_left ← i
    end if
end for
for j ←mid+1 to high do
    sum ← sum+A[j]
    if sum > right_sum then
        right_sum ← sum
        max_right ← j
    end if
end for
return (max_left, max_right, left_sum+right_sum)
```

3.3.2 The Master Theorem

Definition 3.3.2.1: The master theorem is formally defined as follows:
Let $T(n)$ be a function on the natural numbers defined by

$$T(n) = aT\left(\frac{n}{b}\right) + f(n) \tag{21}$$

for some constants $a \geq 1$ and $b > 1$. Then:

1. If $f(n) = O\left(n^{\log_b a - \epsilon}\right)$ for some constant $\epsilon > 0$, then $T(n) = \Theta\left(n^{\log_b a}\right)$.
2. If $f(n) = \Theta\left(n^{\log_b a}\right)$, then $T(n) = \Theta\left(n^{\log_b a} \cdot \log n\right)$.
3. If $f(n) = \Omega\left(n^{\log_b a + \epsilon}\right)$ for some constant $\epsilon > 0$, and if $a \cdot f\left(\frac{n}{b}\right) \leq cf(n)$ for some constant $c < 1$ and for sufficiently large n, then $T(n) = \Theta(f(n))$.

When multiplying two matrices, $A, B \in \mathbb{R}^{n \times n}$ with output $C \in \mathbb{R}^{n \times n}$, we can normally complete the computation in time $\Theta\left(n^3\right)$ by using the following definition:

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} \tag{22}$$

However, we can design a divide-and-conquer algorithm to do this much quicker using MULT(A,B). The running time of MULT(A, B) satisfies

$$T(n) = 8T\left(\frac{n}{2}\right) + \Theta\left(n^2\right) \tag{23}$$

since we have 8 multiplications of $\frac{n}{2} \times \frac{n}{2}$ matrices and 4 matrix additions, each taking time $\Theta\left(\frac{n}{2} \cdot \frac{n}{2}\right) = \Theta\left(n^2\right)$. Therefore, the master theorem implies that $T(n) = \Theta\left(n^3\right)$.

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Algorithm 3 MULT(A,B)

Input: $A, B \in \mathbb{R}^{n \times n}$
Output: $C \in \mathbb{R}^{n \times n}$

Let C be a new $n \times n$ matrix.
if $n = 1$ **then**
  $c_{1,1} \leftarrow a_{1,1} b_{1,1}$
else
  Partition A, B, C into the block matrices.
  $C_{1,1} \leftarrow \text{MULT}(A_{1,1}, B_{1,1}) + \text{MULT}(A_{1,2}, B_{2,1})$
  $C_{1,2} \leftarrow \text{MULT}(A_{1,1}, B_{1,2}) + \text{MULT}(A_{1,2}, B_{2,2})$
  $C_{2,1} \leftarrow \text{MULT}(A_{2,1}, B_{1,1}) + \text{MULT}(A_{2,2}, B_{2,1})$
  $C_{2,2} \leftarrow \text{MULT}(A_{2,1}, B_{1,2}) + \text{MULT}(A_{2,2}, B_{2,2})$
end if
return C

3.3.3 Strassen's Algorithm

Definition 3.3.3.1: Strassen (1969) made a sophisticated divide-and-conquer approach called Strassen's algorithm. It has four steps:

1. Divide A, B, and C into four parts as before, with running time $\Theta(1)$.
2. Create 10 auxiliary matrices $S_1, ..., S_10 \in \mathbb{R}^{\frac{n}{2} \times \frac{n}{2}}$, obtained by sums or differences of $A_{1,1}, ..., B_{2,2}$, with running time $\Theta\left(n^2\right)$.
3. Using all the matrices available, recursively compute seven product matrices $P_1, ..., P_7 \in \mathbb{R}^{\frac{n}{2} \times \frac{n}{2}}$, with running time $7T\left(\frac{n}{2}\right)$.
4. Compute $C_{1,1}, ..., C_{2,2}$ by adding/subtracting combinations of $P_1, ..., P_7$, with running time $\Theta\left(n^2\right)$.

Thus, the worst-case running time of Strassen's algorithm satisfies

$$T(n) \leq 7T\left(\frac{n}{2}\right) + \Theta\left(n^2\right) \tag{24}$$

so, by the master theorem, we obtain

$$T(n) = \Theta\left(n^{log_2 7}\right) = \Theta\left(n^{2.8074}\right) \tag{25}$$

3.3.4 Rod Cutting

Definition 3.3.4.1: The rod cutting problem states that you work for a company that buys steel rods and cuts them into shorter rods, which it then sells. You are asked to determine the best way to cut the rods so as to maximise profit. You know, for $i = 1, ..., n$, the price p_i in euros that the company charges for a rod of length i centimeters. Given a rod of length n and a table of prices p_i for $i = 1, ..., n$, determine the maximum revenue r_n obtained by cutting up the rod and selling the pieces.

Definition 3.3.4.2: An integer partition of a positive integer n is a list of positive integers $(a_1, ..., a_k)$ such that $a_1 \leq ... \leq a_k$ and

$$\sum_{i=1}^{k} a_i = n \tag{26}$$

Let $p(n)$ denote the number of integer partitions of n, then

$$p(1) = 1, p(2) = 2, p(3) = 3, p(4) = 5 \tag{27}$$

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Hardy and Ramanujan proved that

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right) \tag{28}$$

Therefore, brute force has an exponential running time.

Definition 3.3.4.3: Optimal Substructure: Optimal solutions to a problem incorporate optimal solutions to related subproblems, which we may solve independently.

More specifically, for the rod cutting problem, we have for all $n \geq 1$,

$$r_n = \max\{p_i + r_{n-i} : 1 \leq i \leq n\} \tag{29}$$

The idea behind this is that p_n corresponds to no cuts and $p_i + p_{n-i}$ to cutting into a rod of length i and another rod of length $n - i$. To solve the original problem of size n, we solve independent smaller problems of the same type, but of smaller sizes. For the rod problem, we can use the following algorithm:

Algorithm 4 ROD(p, n)

Input: p, n
Output: q

if $n = 0$ **then return** 0
end if
$q \leftarrow -1$
for $i \leftarrow 1$ to n **do**
    $q \leftarrow \max\{q, p[i] + ROD(p, n - i)\}$
end for
return q

3.4 Dynamic Programming Algorithms

3.5 Machinings in Graphs

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Part I
Appendix

4 Bibliography

4.1 Module Professors

Module Lead: Doctor Barnaby Martin

- Focus: Complexity theory, finite model theory, and the links between logic and complexity.
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- Focus: Combinatorics.
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- Focus: Algorithms and networks.
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4.2 References

Corneil, D.G., Perl, Y., and Stewart, L.K. (1985). A Linear Recognition Algorithm for Cographs. SIAM Journal on Computing, 14(4), pp.926-934. doi:10.1137/0214065.

4.3 List of Figures

4.4 Recommended Reading

4.4.1 Algorithms and Complexity

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C. (2009). Introduction to Algorithms. 3rd edn. MIT Press.